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9 / 10 submitted

Q1. Selection Sort

Score: 10.00 TC: 3/3 passed

CODE

```
arr=list(map(int,input().split()))
for i in range(len(arr)):
    min_index=i
    for j in range(i+1,len(arr)):
        if arr[j]<arr[min_index]:
            min_index=j
    arr[i],arr[min_index]=arr[min_index],arr[i]
print(arr)
```

INPUT

23 78 45 8 32 56

OUTPUT

[8, 23, 32, 45, 56, 78]

Q2. Insertion Sort

Score: 10.00 TC: 3/3 passed

CODE

```
arr=list(map(int,input().split()))
for i in range(1,len(arr)):
    key=arr[i]
    j=i-1
    while j>=0 and arr[j]>key:
        arr[j+1]=arr[j]
        j-=1
    arr[j+1]=key
print(arr)
```

INPUT

8 5 7 1 9 3

OUTPUT

[1, 3, 5, 7, 8, 9]

Q3. Merge Sort

Score: 10.00 TC: 3/3 passed

CODE

```
a=list(map(int,input().split()))
def merge_sort(a):
    if len(a)≤1:
        return a
    mid = len(a)//2
    left = merge_sort(a[:mid])
    right = merge_sort(a[mid:])
    result=[]
    i=j=0
    while i<len(left) and j<len(right):
        if left[i]<right[j]:
            result.append(left[i])
            i+=1
        else:
            result.append(right[j])
            j+=1
    result+=left[i:]
    result+=right[j:]
    return result
print(merge_sort(a))
```

INPUT

38 27 43 3 9 82 10

OUTPUT

[3, 9, 10, 27, 38, 43, 82]

Q4. Diagonal Matrix

Score: 10.00 TC: 3/3 passed

CODE

```
s=input()
s=s.replace(" ", "")
s=s[1:-1]
rows=s.split(",")
a=[]
for row in rows:
    row=row.replace("[", "").replace("]", "")
    a.append(list(map(int, row.split(","))))
diagonal= True
for i in range(len(a)):
    for j in range(len(a[i])):
        if i≠j and a[i][j]≠0:
            diagonal= False
if diagonal:
    print("Diagonal Matrix")
else:
    print("Not Diagonal Matrix")
```

INPUT

[[1,0,0],[0,5,0],[0,0,8]]

OUTPUT

Diagonal Matrix

Q5. Row and Column Sums

Score: 0.00 TC: 0/3 passed

CODE

```
s=input().strip()
s=s.replace(" ", "")
s=s[1:-1]
rows=s.split(",")
a=[]
for row in rows:
    row=row.replace("[", "").replace("]", "")
    a.append([int(x) for x in row.split(",")])
row_sum=[sum(r) for r in a]
col_sum=[sum(a[i][j] for i in range(len(a))) for j in range(len(a[0]))]
print("Row=", row_sum)
print("col=", col_sum)
```

INPUT

[[1,2,3],[4,5,6]]

OUTPUT

Row= [6, 15]
col= [5, 7, 9]

Q6. Maximum Subarray Sum

Score: 6.67 TC: 2/3 passed

CODE

```
a=list(map(int,input().split()))
best=a[0]
for i in range(len(a)):
    total=0
    for j in range(i,len(a)):
        total=total+a[j]
        if total>best:
            best=total
print(best)
```

INPUT

-1 -2 -3

OUTPUT

-1

Q7. Common Elements Among Three Lists

Score: 10.00 TC: 3/3 passed

CODE

```
s=input()
parts=s.split(";")
a=list(map(int,parts[0].split()))
b=list(map(int,parts[1].split()))
c=list(map(int,parts[2].split()))
common=[]
for x in a:
    if x in b and x in c and x not in common:
        common.append(x)
print(common)
```

INPUT

1 2 3;2 3 4;2 5

OUTPUT

[2]

Q8. Matrix Multiplication

Score: 6.67 TC: 2/3 passed

CODE

```
s=input().replace(" ","")
parts=s.split("B=")
A=parts[0].replace("A=", "")
B=parts[1]
A=A[2:-2]
B=B[2:-2]
matrixA=[]
matrixB=[]
for row in A.split(","):
    matrixA.append(list(map(int,row.split(","))))
for row in B.split(","):
    matrixB.append(list(map(int,row.split(","))))
result=[]
for i in range(len(matrixA)):
    row=[]
    for j in range(len(matrixB[0])):
        total=0
        for k in range(len(matrixB)):
            total+=matrixA[i][k]*matrixB[k][j]
        row.append(total)
    result.append(row)
for i,row in enumerate(result):
    if i==0:
        print("["+str(row), end="")
    else:
        print(str(row), end="")
    print("]" if i==len(result)-1 else "")
```

INPUT

A=[[1,2],[3,4]] B=[[5,6],[7,8]]

OUTPUT[[19, 22]
[43, 50]]**Q9. Sort Rotate Min Max**

Score: 3.33 TC: 1/3 passed

CODE

```
s=input()
s=s.replace("List=", "").replace("k=", "")
p=s.split(",")
L=list(map(int,p[0].split()))
k=int(p[1])
L.sort()
R=L[k:]+L[:k]
print("Sorted=",L)
print("Rotated=",R)
print("Min=",min(L))
print("Max=",max(L))
```

INPUT

List=5 2 8 1 4, k=2

OUTPUTSorted= [1, 2, 4, 5, 8]
Rotated= [4, 5, 8, 1, 2]
Min= 1
Max= 8**Q10. Sort Matrix Search**

— Not Submitted —